Program 10: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Step 1: Start the minkube and check status

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ minikube start --cpus=2 --memory=4096
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
❗  You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
    ▪ Using image registry.k8s.io/ingress-nginx/controller:v1.13.2
    ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
    ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.2
🔎  Verifying ingress addon...
🌟  Enabled addons: ingress, default-storageclass, storage-provisioner, metrics-server

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
    ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check status of minikube

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create deployments.yaml

```
  GNU nano 7.2                                                   deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: Create services.yaml

```
  GNU nano 7.2
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

Step 5:Apply those

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 6 : Get the pods created

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ kubectl get pods
NAME                                READY   STATUS      RESTARTS        AGE
nginx-deployment-6f9664446b-5m82f   1/1     Running     2 (2m41s ago)   7d
nginx-deployment-6f9664446b-krd2k   1/1     Running     2 (2m41s ago)   7d
nodejs-pod                          1/1     Running     2 (2m31s ago)   7d
test-cron-29435303-48bst            0/1     Completed   0               104s
test-cron-29435304-k5hcc            0/1     Completed   0               82s
test-cron-29435305-wkdxd            0/1     Completed   0               22s
```

Step 7 : Check for the minikube dashboard

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:43175/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Opening in existing browser session.
```

Step 8 : Expose the service through minikube

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/p10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30483

🎉 Opening service default/nginx-service in default browser...

OUTPUT :